CompSci 234/NetSys 210
Advanced Topics in Networking
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Unit 6: Video Compression Overview

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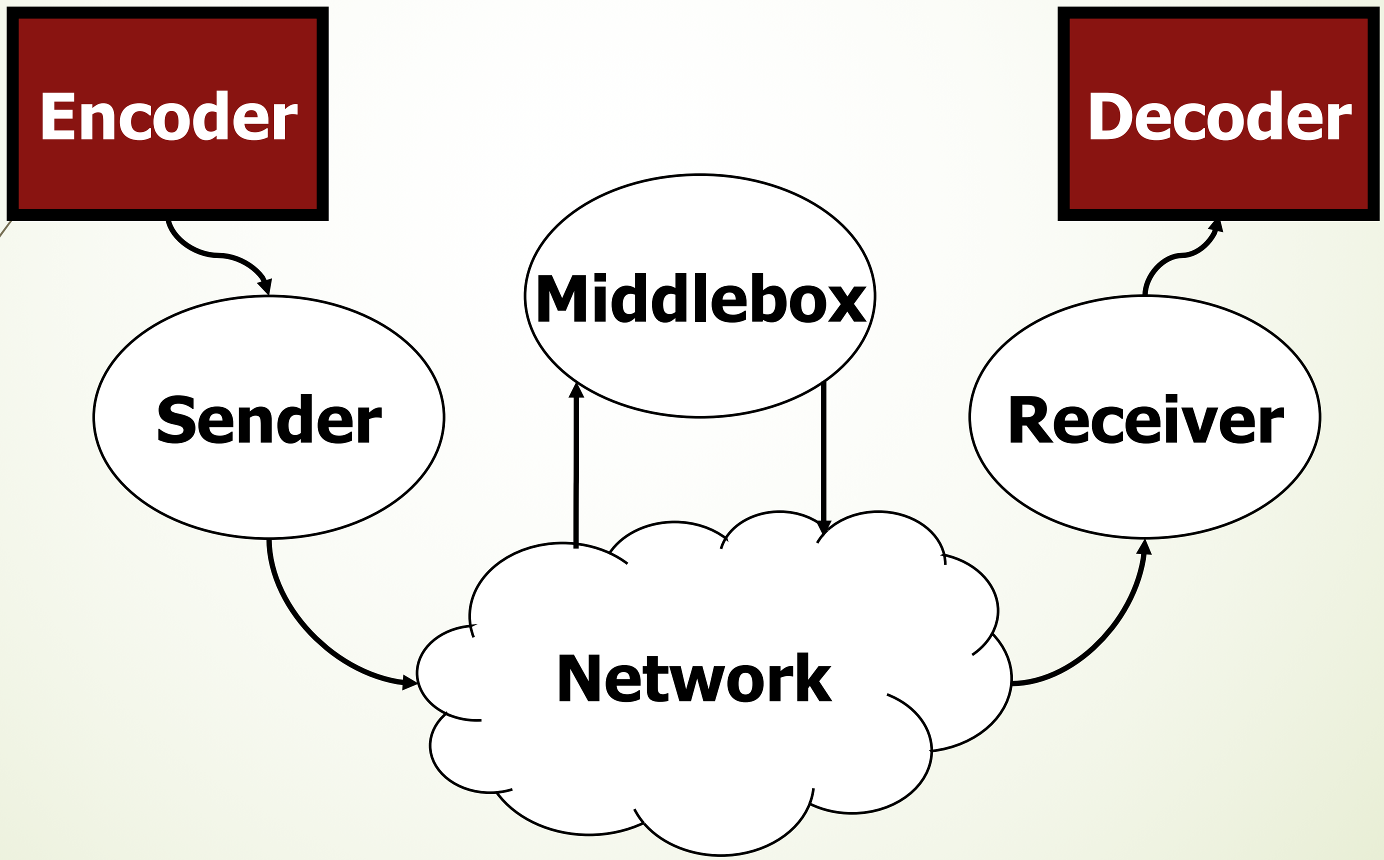
Slide adopted from Profs. Ooi and Zimmerman, and Ross' materials

Agenda

- Why Compression
- Image Coding Tools
- Video Coding Tools
- Popular Video Codecs

We are Here



Why Compression

- "Bandwidth is Not Enough"
- "Disk Space is Not Enough"

- Size of Uncompressed DVD Movie = (720 x 576) pixels x 3 bytes x 25 fps x 60 sec/min x 120 min = 208.6 GB
- NTSC: 29.97 fps (30/1.001); PAL 25 fps

Optical Disc Formats (1)

- CD: ~650 MB
  - Video CD: codec MPEG-1
  - 1X max. read speed: 1.5 Mb/s
- DVD:
  - 4.7 (4.38) GB (single layer)
  - 8.5 (7.92) GB (dual layer)
  - Double layer and dual sided (up to 18 GB)
  - 1X max. read speed: ~10 Mb/s
  - Video codec: MPEG-2

Optical Disc Formats (2)

- Blu-ray
  - Capacity: 25 GB and 50 GB
  - 1X speed: 36 Mb/s ← Still too slow
  - Video codec: VC-1, H.264, MPEG-2
- Solution: Lossy compression

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Core Idea of Image Compression

- Throw away information **we cannot see**, i.e., based on human visual systems:
  - Color information: we are more sensitive to luminance info.
  - High frequency signals
- Rearrange data for good compression
- Use common compression tools
  - DCT -> Quantizer -> Entropy Coding

Controllable Compression Ratio

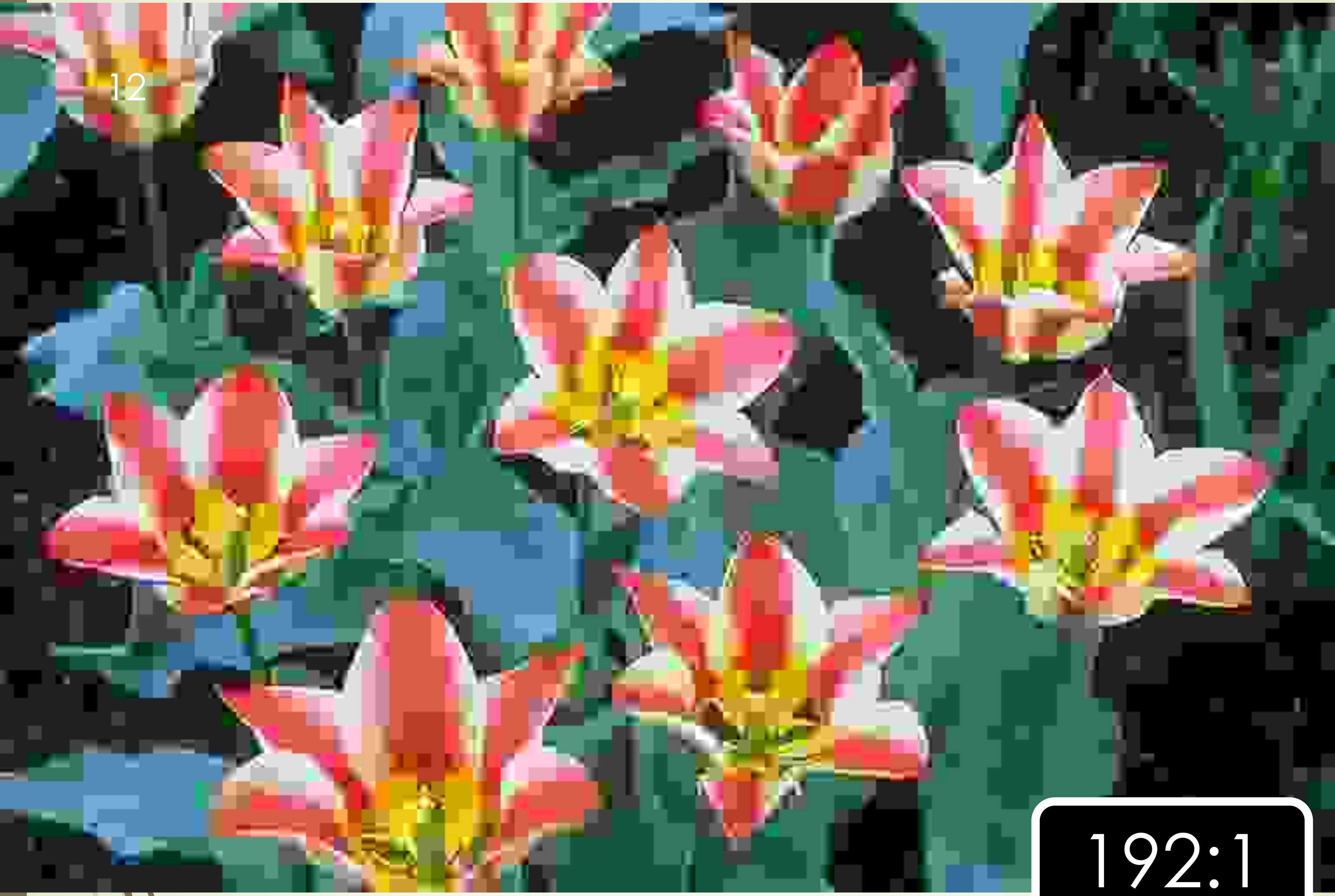
Quality	Size	Ratio
Raw TIFF	1153KB	1:1 **No Compression**
Zipped TIFF	982KB	1.2:1 **Lossless**
Q=100	331KB	3.5:1
Q=70	67KB	17:1
Q=40	43KB	27:1
Q=10	16KB	72:1
Q=1	6KB	192:1

JPEG

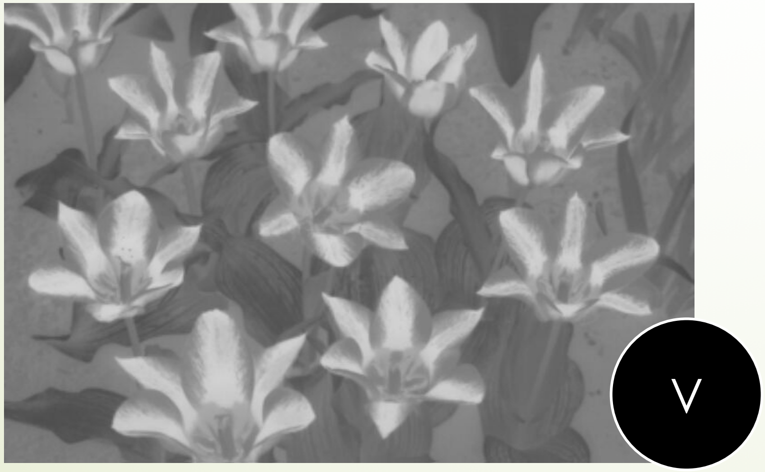
Lossy

1:1

27:1


192:1

(Discard) Color Information



Color Subsampling

- The subsampling scheme is commonly expressed as a three part ratio (e.g. 4:2:2). The parts are (in their respective order):
- **Luma** (Y) horizontal sampling reference (originally, as a multiple of 3.579 MHz in the NTSC television system).
- **Cr** (U) horizontal factor (relative to first digit).
- **Cb** (V) horizontal factor (relative to first digit), except when zero. Zero indicates that **Cb** horizontal factor is equal to second digit, and, in addition, both **Cr** and **Cb** are subsampled 2:1 *vertically*. Zero is chosen for the bandwidth calculation formula to remain correct.
- To calculate required bandwidth factor relative to 4:4:4, one needs to sum all the factors and divide the result by 12.